A minimum Venn diagram of order eight

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Certificates, checkers and Lean proof: `https://github.com/dzoba/minimum-venn-diagram-8`

Abstract

Bultena and Ruskey proved in 1998 that an n-Venn diagram has at least $L_n = \lceil (2^n-2)/(n-1) \rceil$ crossings and conjectured that the bound is attained for every $n \geq 8$. We give an 8-Venn diagram with $37 = L_8$ crossings, which settles the case $n = 8$ and answers a question of Brenner, Gregor, Mütze and Verciani; the best known before had 39 crossings (Seamons). Its 37 crossings are listed in an appendix, and it has been checked by three separately written programs (see the section on the use of AI tools) and in Lean 4. It has a half-turn symmetry, and in every 8-Venn diagram with 37 crossings a finite group of orientation-preserving symmetries has order at most 2. Doubling gives a 9-Venn diagram with 74 crossings, optimal among those in which some curve meets every other curve exactly twice.

1 Introduction

An *n-Venn diagram* is a collection of n simple closed curves in the plane that intersect in finitely many points and create exactly 2^n connected regions, one for each combination of being inside or outside each curve. In any intersection point of at least two curves at least two of them must cross, and such a point is called a *crossing* [1]. Several curves may pass through one crossing, and some of them may only touch there. Bultena and Ruskey [2] proved that an n-Venn diagram, $n \geq 2$, has at least $L_n = \lceil (2^n - 2)/(n - 1) \rceil$ crossings, called diagrams that attain this bound *minimum*, and conjectured that minimum n-Venn diagrams exist for all $n \geq 8$ (see also [7]). They are known for $n \leq 7$ [2, 1]. For $n = 8$, $L_8 = 37$. Brenner, Gregor, Mütze and Verciani [1] constructed an 8-Venn diagram with 40 crossings and asked whether 37 is possible. Seamons [8] reduced 40 to 39 by an edge surgery on a reconstruction of their diagram.

Theorem 1. *There is an 8-Venn diagram with 37 crossings. So the least number of crossings of an 8-Venn diagram is 37, and the conjecture of Bultena and Ruskey holds for $n = 8$.*

The proof is a certificate, the dual graph of the diagram, given by its 37 faces in Appendix A. The certificate, three checkers and their logs, and a Lean proof are at `https://github.com/dzoba/minimum-venn-diagram-8`.

2 The certificate

We use dual graphs as in [1, Section 1.3]. Let Q_n be the graph on the subsets of $[n] = \{1, \ldots, n\}$ in which two sets are adjacent when they differ in one element, the *direction* of the edge. The dual $Q(D)$ of an n-Venn diagram D has a vertex for each region, the set of curves containing it, and an edge across each arc of a curve. It satisfies: (1) it is a spanning subgraph of Q_n; (2) it is a plane graph in which each face has length 2ℓ for some $2 \leq \ell \leq n$ and contains exactly two edges of each of ℓ distinct directions; (3) for every $j \in [n]$ the subgraphs induced by the vertices x with $j \in x$ and with $j \notin x$ are connected. Conversely, the dual of any subgraph of Q_n with these properties is an n-Venn diagram [1]. The faces of $Q(D)$ are the crossings of D. We work on the sphere; removing a point of a region gives a diagram in the plane.

Theorem 1 rests on the converse, which [1] state without proof, so we give the argument. Let H be a connected plane graph satisfying (1)–(3), and let H^* be its plane dual: a point in each face of H, and for each edge e an arc e^* that crosses e once and joins the points of the faces on its two sides. For $j \in [n]$ let $S_j = \{x : j \in x\}$. The edges of direction j form the cut $\delta(S_j)$, and by (3) both of its

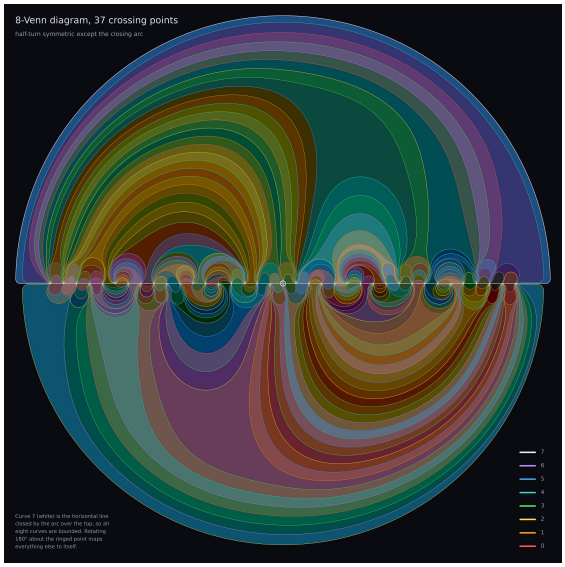


Figure 1: The diagram of `venn8_37_halfturn_a.json`, drawn and checked against the certificate by `render37.py`. Curve j of the legend is curve $j+1$ in the text; curve 8 (white) is the line closed by the arc over the top. Rotation by 180° about the ringed crossing preserves the drawing except along that arc.

sides induce connected subgraphs, so $\delta(S_j)$ is a bond (a minimal nonempty cut) of H. As H is the plane dual of H^*, the arcs e^* with $e \in \delta(S_j)$ form a cycle of H^* [4, Proposition 4.6.1]; this cycle, a simple closed curve, is curve j. The other edges of H do not meet it and each edge of $\delta(S_j)$ crosses it once, so S_j lies on one side of curve j and the other vertices on the other side. Distinct curves consist of distinct arcs, so they meet only in the points of the faces, finitely many. The union of the curves is H^*, whose faces are the regions, one around each vertex x of H. Taking the side of curve j that contains S_j as its inside (the bounded side once a point of the region $\emptyset$ is removed), the region of x lies inside curve j exactly when $j \in x$. By (1) there are 2^n regions, one for each combination.

A certificate lists the faces of a plane graph, each as the cyclic sequence of vertex labels around it, with $x \subseteq [n]$ stored as $\sum_{i \in x} 2^{i-1}$; so curve i is bit $i-1$ in the files, the Lean code and the legend of Figure 1. A face list in which every edge is traversed once in each direction, the faces around every vertex form a single cycle, the graph is connected and $V - E + F = 2$ is a combinatorial map of the sphere, and it determines a plane graph with exactly these faces, unique up to homeomorphism [5, Section 3.2]. The certificate `venn8_37_halfturn_a.json` (SHA-256 `7c1fed92...c41dd12`) has 256 vertices, 291 edges and 37 faces, 32 of length 16 and five of length 14. So 32 crossings lie on all eight curves and five on seven. Curves 1, 3, 5, 7 and 8 pass through all 37 crossings, curve 4 through 36, curves 2 and 6 through 35. At every crossing some pair of curves crosses.

3 The lower bound and the checks

We recall why 37 is least [2]. The dual of an n-Venn diagram is a connected plane graph with $V = 2^n$ vertices, E edges and F faces, so $V - E + F = 2$. Face lengths are at most $2n$ and sum to $2E$, so $E \le nF$ and $2^n - 2 = E - F \le (n-1)F$, hence $F \ge L_n$. For $n = 8$, $F \ge \lceil 254/7 \rceil = 37$, and $F = 37$ forces $E = 291$ and $\sum_f (8 - \ell_f) = 5$, where $2\ell_f$ is the length of face f.

The repository `https://github.com/dzoba/minimum-venn-diagram-8` contains the certificates, three checkers and their logs, the Lean project and the drawing program. Theorem 1 reduces to checking that 37 faces satisfy (1)–(3) and form a plane graph. Three separately written checkers do this in a fraction of a second: `check_a.py` checks (1)–(3), the induced rotation system and Euler's

formula; `check_b.py` builds the primal diagram and checks that the arcs of each curve form one closed curve; `ref37_v2.py`, written from scratch by a separate Claude instance acting as referee, also checks that some pair of curves crosses at each crossing. All three accept both 37-crossing certificates, Seamons's 39 and his reconstruction of the 40 of [1]. They reject each of these with a face removed, and two small inputs, built by the referee of this release, in each of which exactly one condition fails: condition (3), or the vertex links (`python3 checkers/run_all.py`).

For the face list `cert` of `venn8_37_halfturn_a.json`, a Lean 4 proof (with Mathlib) in `lean/` establishes

$$\texttt{(IsVennDual8 cert} \wedge \texttt{CrossingAtEachPoint cert)} \wedge \texttt{cert.length = 37} \wedge \forall\, \texttt{fs, IsVennDual8 fs} \to 37 \le \texttt{fs.length}.$$

`IsVennDual8` states (1)–(3) for an oriented face list and that the list is a combinatorial map of the sphere (each edge traversed once in each direction, vertex links single cycles, connected, $V-E+F=2$); `CrossingAtEachPoint` says that two directions interleave on every face. The facts about `cert` are proved by kernel evaluation (`decide +kernel`), the bound by a general argument, using only the axioms `propext`, `Classical.choice` and `Quot.sound`. The kernel evaluation, in `CertProof.lean`, needs about 16 GiB of memory. Codex compiled the whole project from the same `.lean` files under a different Lake configuration; the axiom list in `lean/AXIOMS.txt` is the output of that build, whose log is not in the repository. With the Lake configuration as released we built `Defs`, `Cert`, `Decide`, `LowerBound` and `Sound`, every module except `CertProof` and the two that depend on it, and confirmed that the bound `faces_ge_37` uses only the three axioms above. The forward direction of the characterization of [1] and the passage from a combinatorial map to a drawing are cited, not formalized.

4 Symmetry and a second diagram

A symmetry of a diagram on the sphere is a homeomorphism mapping the set of curves onto itself. Every diagram has infinitely many, for example any homeomorphism that is the identity outside a small disc in one region, so we count them by their action on labels. A symmetry acts by $x \mapsto \pi(x) \triangle c$ for a permutation π of $[n]$ and a set $c \subseteq [n]$, and this label map sends the face list onto itself, preserving the orientation of every face or reversing all of them. We call such label maps *automorphisms* of the diagram. Conversely, a group A of automorphisms is realized by a finite group of homeomorphisms isomorphic to A, acting on the drawing that the face list determines, which is homeomorphic to the diagram: each automorphism permutes the triangles of the barycentric subdivision of the decomposition into crossings, arcs and regions, and we map each triangle affinely onto its image. So Proposition 2 also shows that a diagram with 37 crossings has at most two orientation-preserving automorphisms.

With $\sigma = (1\,7)(2\,6)(3\,5)$ and $c = [8] \setminus \{4\}$ (the integer 247), $g(x) = \sigma(x) \triangle c$ maps every face of `venn8_37_halfturn_a.json` to a face with the same orientation: a half-turn. It fixes no region, one arc (dual to the edge between $\{5,6,7\}$ and $\{5,6,7,8\}$, on curve 8) and one crossing (face 21 in Appendix A), on all curves but curve 4. A search over all maps $x \mapsto \pi(x) \triangle c$ finds no other automorphism of the face list, including orientation-reversing ones, so the automorphism group of this diagram has order 2.

Proposition 2. *Let G be a finite group of orientation-preserving symmetries of an 8-Venn diagram with 37 crossings. Then $|G| \le 2$, and a symmetry of order 2 fixes exactly one crossing, one arc and no region.*

Proof. The 37 crossings, 291 arcs and 256 regions are the cells of a decomposition of the sphere. Let $g \in G$ have order $k \ge 2$. By Kerékjártó's theorem [3] g is conjugate to a rotation, whose two poles are the only fixed points of each nontrivial power of g. A cell mapped to itself by such a power

contains a fixed point of it (for an arc since a periodic map of an interval has one, for a region by [3] on its one-point compactification), so the cells without a pole lie in orbits of size k. If v, e, f count the regions, arcs and crossings containing a pole, then $v + e + f \leq 2$ and $256 \equiv v$, $291 \equiv e$, $37 \equiv f \pmod{k}$. If $k \geq 3$ then $e = 0$: an arc containing a pole would be mapped to itself, and as a periodic map of an interval has order at most 2, the nontrivial g^2 would fix the whole arc. So $k \mid 291 = 3 \cdot 97$. For $k \in \{97, 291\}$, $256 \bmod k \in \{62, 256\}$ exceeds $v \leq 2$. For $k = 3$, $v = 1$; but the fixed labels, solutions of $\pi(x) \triangle c = x$, form a coset of a subspace of $\mathbb{F}_2^8$ of dimension the number of cycles of π, so there are 0 or at least 2. Hence $k = 2$, and parity gives $(v, e, f) = (0, 1, 1)$. So G is an elementary abelian 2-group. Its orbits on the 37 crossings have size a power of 2, so it fixes a crossing, and it acts faithfully by rotations on the cyclic sequence of regions around it, since no nontrivial element fixes a region. So G is cyclic and $|G| \leq 2$. □

Remark 3. We sketch why a finite group of symmetries of an 8-Venn diagram with 37 crossings has no orientation-reversing element, so that it has order at most 2; this is not formalized, and for the two diagrams here the search above settles it directly. Let h reverse orientation. As h^2 preserves it, h has order 2 or 4 by Proposition 2. Let h be an involution. It permutes the 37 crossings, an odd number, so it fixes one, and by [3] it is conjugate to a reflection, with a circle M of fixed points (the antipodal map fixes no point). Its permutation π of the curves is an involution and fixes an even number s of curves. If h keeps the two sides of a fixed curve D, it acts on D as a reflection of a circle, so D meets M in exactly two points. A fixed curve with its sides exchanged would be M itself; then $s \geq 2$, and another fixed curve would meet it only twice, whereas any two curves share at least 32 crossings, since by $\sum_f (8 - \ell_f) = 5$ at most five crossings miss a curve. So no arc lies in M, every invariant arc is reversed, and as h has degree -1 its Lefschetz number, computed from the cells [6, Section 22], gives $f + a - v = 0$ for the numbers of invariant crossings, arcs and regions. Here f is odd; $a \leq 2s$, since an invariant arc lies on a fixed curve and contains one of its two points on M; $v \in \{0, 2^{(8+s)/2}\}$, the number of solutions of $\pi(x) \triangle c = x$; and $f \leq 7$ if $s > 0$, since the 32 crossings on all eight curves lie on a fixed curve and at most two of them on M. No case fits: for $s = 0$, $a = 0$ and $f = v$ is even, and for $s > 0$, $f = v - a$ is at most 0 or at least $32 - 16$. If h had order 4, then h^2 would be the half-turn of Proposition 2, h would map its unique invariant arc onto itself, and h^2, the square of a homeomorphism of that arc, would preserve its ends and, being periodic, fix it pointwise; but a half-turn has only two fixed points.

The certificate `venn8_37_halfturn_b.json` is a second diagram with 37 crossings, with the same face lengths and symmetry g and, by the same search, no other automorphism. It is not isomorphic to the first, even allowing relabelling of curves, exchange of inside and outside, and reflection: the numbers of crossings on the curves are $34, 36, 36, 37, 37, 37, 37, 37$ against $35, 35, 36, 37, 37, 37, 37, 37$, and the dual graphs have 196, 50, 10 vertices of degree 2, 3, 4 against 192, 58, 6.

5 Nine and more curves

A face of the dual is *colorful* if it has length $2n$ and contains complementary vertices x and $[n] \setminus x$. By [1, Lemma 10] (Lemma 11 in arXiv:2511.09230v1, which contains its proof), a diagram whose dual has a colorful face gives an $(n + 1)$-Venn diagram with twice as many crossings and again a colorful face: two copies of the dual, the second with $n + 1$ added to every label, are joined by the edges $\{x, x \cup \{n + 1\}\}$ and $\{\bar{x}, \bar{x} \cup \{n + 1\}\}$ for complementary $x, \bar{x}$ on that face. The face of `venn8_37_halfturn_a.json` containing $\{6, 7, 8\}$ and $\{1, \ldots, 5\}$ (face 1 in Appendix A) is colorful, which gives n-Venn diagrams with $37 \cdot 2^{n-8}$ crossings (Table 1; for $n = 16$ the 5118 of [1] is smaller). The files for $9 \leq n \leq 15$ were checked by the three programs in `checkers/`; `checkers/invariants.py` confirms that each is the doubling of the previous one.

In `venn9_74.json` curve 9 meets every other curve in exactly two points. In this class 74 is optimal.

n	8	9	10	11	12	13	14	15
lower bound L_n [2]	37	64	114	205	373	683	1261	2341
[1, Theorem 2]	40	80	160	320	640	1280	2560	5120
this note	37	74	148	296	592	1184	2368	4736

Table 1: Numbers of crossings; for $n \geq 9$ the last row is by doubling (Seamons's 39 gives 78).

Proposition 4. *Let $n \geq 3$, $m = n-1$, and let an n-Venn diagram with F crossings have a curve c that meets every other curve in exactly two points. If k crossings lie on c, then $F \geq 2(L_m - 1) + k \geq 2L_m$; for $n = 9$, $F \geq 74$.*

Proof. Another curve j meets c in two points p, q, and has points on both sides of c, so one of its arcs between p and q lies inside c and the other outside. Let Δ be the closed disc bounded by c on one side, and collapse c to a point P. The other m curves become simple closed curves through P, whose union is a connected plane graph with vertices P and the I crossings inside Δ, all of degree at most $2m$, and with the 2^m regions inside Δ as faces. Euler's formula gives $2^m - 2 = E - (I+1) \leq (m-1)(I+1)$, so $I + 1 \geq L_m$. The same holds on the other side, and $k \geq 2$ crossings lie on c, so $F \geq 2(L_m - 1) + k$. □

This bounds one class only; in general only $L_9 = 64$ is known. Our search also found 74-crossing 9-Venn diagrams outside the class (`venn9_74_mixed_r2.json`, `venn9_74_k3_nearmiss.json`). Local searches from these and other diagrams with 74 and 75 crossings found none with 73; they were not exhaustive and give no bound.

How the diagrams were found. A Metropolis walk over plane spanning subgraphs of Q_8, given as rotation systems, allowed unbalanced faces and disconnected curve halves at a cost and applied each move together with its image under g. From half-turn invariant arrangements of the nested cycles of [1] it produced many diagrams with 38 and 39 crossings, and walks from two 39s with a higher cost per edge reached the two 37s. The search programs are not included.

Data availability

The certificates, the checkers and their logs, the negative controls, the Lean project and the drawing program are at `https://github.com/dzoba/minimum-venn-diagram-8`. The face list of Theorem 1 is also printed in Appendix A.

Use of AI tools

This work was carried out with AI systems under the author's direction. Claude (Anthropic) wrote and ran the searches, the checkers `check_a.py` and `check_b.py` and the other scripts in the repository, the doubling and the drawing, wrote the Lean formalization, and drafted the text. Separate Claude instances wrote `ref37_v2.py` and refereed this release; the referee built the two targeted negative controls. The proofs of Sections 4 and 5 were worked out by Claude and Codex (OpenAI). Codex contributed the exclusion of reflections and reviewed Proposition 4. It also wrote two further checkers and a reconstruction of the first diagram, which accept both 37-crossing diagrams and the 74-crossing ones, and compiled and reviewed the Lean proof; these checkers, the reconstruction and the build logs are not part of the repository. The author is responsible for the content.

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A The 37 faces

The faces of `venn8_37_halfturn_a.json`, numbered from 0 in the order of the file (as in the logs), with the orientation of the file. Row k is the cyclic sequence of region labels around crossing k; a label is $\sum_{i\in x} 2^{i-1}$ for the set x of curves containing the region, and consecutive labels, the last and the first included, are the two sides of an arc. Faces 2, 5, 21, 25 and 26 have length 14 and miss curves 6, 2, 4, 2 and 6. Face 1 is the colorful face of Section 5, and face 21 is the crossing fixed by the half-turn g. The conditions of Section 2 can be checked from this table alone.

0	0	1	3	11	27	91	123	107	75	67	71	69	68	196	192	128
1	0	128	192	224	225	241	209	145	17	19	23	31	15	7	3	1
2	2	6	4	5	133	129	193	65	81	83	82	90	26	10		
3	2	10	26	58	122	250	251	255	253	245	229	197	133	5	4	6
4	3	7	15	47	175	171	235	234	232	248	249	121	89	91	27	11
5	8	24	88	216	217	221	93	77	109	101	97	105	104	72		
6	8	40	32	36	38	39	167	231	199	215	223	219	217	216	88	24
7	8	72	104	120	124	92	220	222	214	198	206	202	200	201	137	136
8	8	136	137	201	73	9	41	57	49	53	55	39	38	36	32	40
9	9	73	201	200	202	206	198	214	246	182	54	55	53	49	57	41
10	12	44	108	100	96	98	114	115	243	211	195	203	207	205	204	76
11	12	76	204	205	207	203	195	211	243	242	226	194	130	138	142	140
12	12	140	141	143	139	131	163	227	99	115	114	98	96	100	108	44
13	13	29	28	92	124	120	56	184	186	154	155	159	151	135	143	141
14	13	141	140	142	138	130	162	178	182	246	214	222	220	92	28	29
15	14	30	22	18	16	80	84	86	87	95	79	111	239	238	110	46
16	14	46	110	238	239	237	236	228	244	212	208	144	16	18	22	30
17	15	31	23	21	20	148	144	208	212	244	228	236	237	239	111	47
18	16	144	176	160	168	172	173	165	181	149	213	85	87	86	84	80
19	17	145	209	241	225	233	169	168	160	176	144	148	20	21	23	19
20	25	89	121	249	248	252	188	180	164	132	134	150	158	156	157	153
21	25	153	157	156	152	154	186	187	59	43	107	123	91	89		
22	26	90	82	83	81	85	213	149	181	165	173	172	174	170	42	58
23	33	35	34	162	130	194	226	242	243	115	113	117	125	109	45	37
24	33	37	45	109	77	93	221	217	219	218	210	146	178	162	34	35
25	39	55	54	182	178	146	210	218	219	223	215	199	231	167		
26	42	170	174	172	164	180	188	252	254	255	251	250	122	58		
27	43	59	187	186	184	56	120	104	105	97	101	103	71	67	75	107
28	47	111	79	95	87	85	81	65	64	66	74	106	234	235	171	175
29	48	50	51	179	147	131	139	143	135	151	183	247	119	118	116	52
30	48	52	60	61	125	117	113	115	99	227	163	131	147	179	51	50
31	52	116	112	240	224	192	196	68	69	71	103	101	109	125	61	60
32	62	63	127	255	254	252	248	232	234	106	74	66	70	78	94	126
33	62	126	94	78	70	66	64	65	193	129	161	177	185	189	191	190
34	62	190	191	189	185	177	161	129	133	197	229	245	253	255	127	63
35	102	118	119	247	183	151	159	155	154	152	156	158	150	134	166	230
36	102	230	166	134	132	164	172	168	169	233	225	224	240	112	116	118